



Semester 1 Examinations 2013/2014

Exam Code(s)

Exam(s)

Higher Diploma in Software Design and Development

Module Code(s)

CT537

Module(s)

Software Engineering Methods

Paper No.

1

Repeat Paper

External Examiner(s)

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Internal Examiner(s)

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Instructions:

Candidates should answer **Question 1** and **another two** questions.

All questions carry equal marks.

Duration

2 hours

No. of Pages

4

Requirements:

MCQ

Handout

Statistical/ Log Tables

Cambridge Tables

Graph Paper

Log Graph Paper

Other Materials

Release to Library: Yes ☐

No ☐

1. Draw a **context diagram**, a **systems level DFD** and one **third level DFD** to represent the data and processing involved in the following system description for an online **university registration system**:

The system should enable the staff of each academic department to examine the courses offered by their department, add and remove courses, and change the information about them (e.g., the maximum number of students permitted). It should permit students to examine currently available courses, add and drop courses to and from their schedules, and examine the courses for which they are enrolled. Department staff should be able to print a variety of reports about the courses and the students enrolled in them. The system should ensure that no student takes too many courses and that students who have any unpaid fees are not permitted to register.

(16)

(b) You have been asked to select an appropriate **process model** for the development of the above project (online university registration system). Evaluate your alternatives and select one, clearly outlining your evaluation criteria and the reasons for your selection.

(4)

2. (a) In software development projects, deciding “**what**” to say is recognised as being more difficult than “**how**” to say it. Recent trends have further complicated this process for developers. Discuss what requirements processes and techniques you would use as a project manager to handle the following requirements challenges, illustrating your response with appropriate examples in each case:

- continuously evolving requirements
- short product cycles

(8)

(b) Construct a structured specification for the delivery charging process of FAST Courier Service described below. Explain why you chose the particular specification approach you used.

If the order is worth £2000 or more, the charge is 1% of order value for deliveries within a 80 KM radius, and 1.75% for those on or outside this radius. For orders below £2000, the charge is 1.75% within an 80-KM radius, and 2.5% for those on or outside this radius. If express delivery is required, the above rates are doubled for deliveries outside an 80-KM radius.

(6)

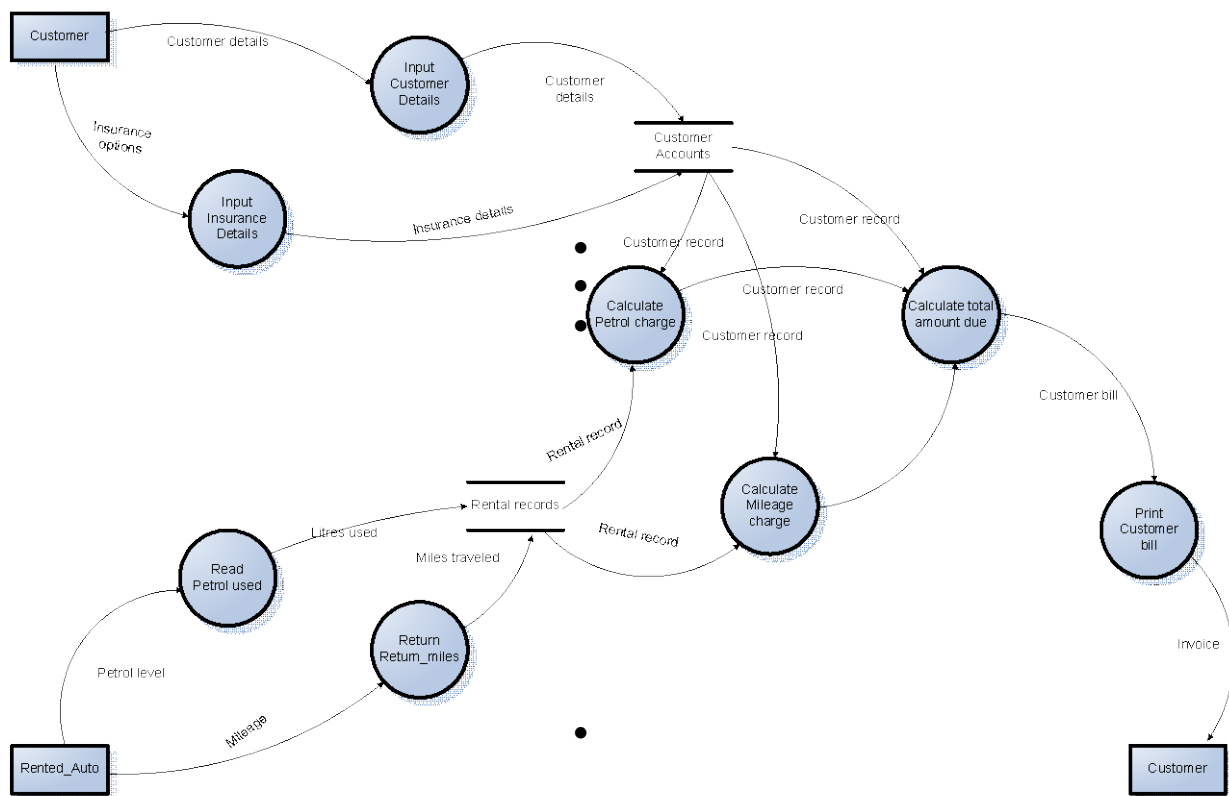
(c) Provide a brief report detailing how the **implementation** of the Arts Festival system in Q.1 will be approached. The report should detail the design method, programming language and framework employed, stating all assumptions made.

(6)

3. (a) Construct a structure chart from the DFD shown below, using transform and transaction analysis. Your chart should show all relevant data couples and flags. Refine your first-cut structure chart as necessary, applying the design criteria of coupling and cohesion.

(10)

KoolKars Automobile Rentals



(b) Elaborate on the role and contribution of modelling in the software development lifecycle. **(4)**

(c) In assessing the quality of a proposed design, software designers examine the coupling, cohesion and overall factoring of their solution. What are the guidelines which exist for the designer in this regard?

Identify the types of coupling evident in each of the following interfaces from the calling module to a called module that prints letters:

- (a) CALL PRINT LABELS USING TITLE, NAME, ADDRESS
 - (b) CALL PRINT LABELS USING CLIENT_DETAILS
 - (c) CALL PRINT LABELS USING VALID_FLAG, CLIENT_DETAILS
- (6)**

4. (a) Prepare a testing strategy or plan for a system of your choice (e.g. Sports Club System, Student Registration system etc.). Your strategy should incorporate both low and high order testing, and should be illustrated with practical examples of testing techniques where appropriate. **(8)**

(b) What are the basic techniques use to estimate the size of a project? What trade-offs must a project manager manage to deliver a project to schedule? **(4)**

(c) You have been appointed software project manager for a new games development company. Your job is to build a breakthrough product that combines a new gaming console with state of the art software. Because competition for the home entertainment market is intense, there is significant pressure to get the job done.

- (i) Make a prioritised list of technology risks that you would face as you begin a project of this type.
 - (ii) Provide contingency plans for each risk listed.
- (8)**